

22 A compound, Z, decomposes by a first order reaction.

At the start of the reaction, the concentration of Z is $0.400 \text{ mol dm}^{-3}$.

After 600 s, the concentration of Z falls to $0.100 \text{ mol dm}^{-3}$.

After which total time would the concentration of Z become $0.050 \text{ mol dm}^{-3}$?

A 700 s

B 750 s

C 900 s

D 1200 s